From Linear Regression to the Artificial Neuron

Deep Learning

Daniel E. Acuna
Associate Professor, University of Colorado Boulder

Contents of This Video

In this video, we will cover:

- Review of linear regression as a foundation
- The perceptron: the simplest artificial neuron
- Mathematical formulation of a neuron
- Connecting neurons to logistic regression
- Why this matters for deep learning

Linear Regression: Our Starting Point

$$\hat{y} = w_1 x_1 + w_2 x_2 + \cdots + w_n x_n + b$$

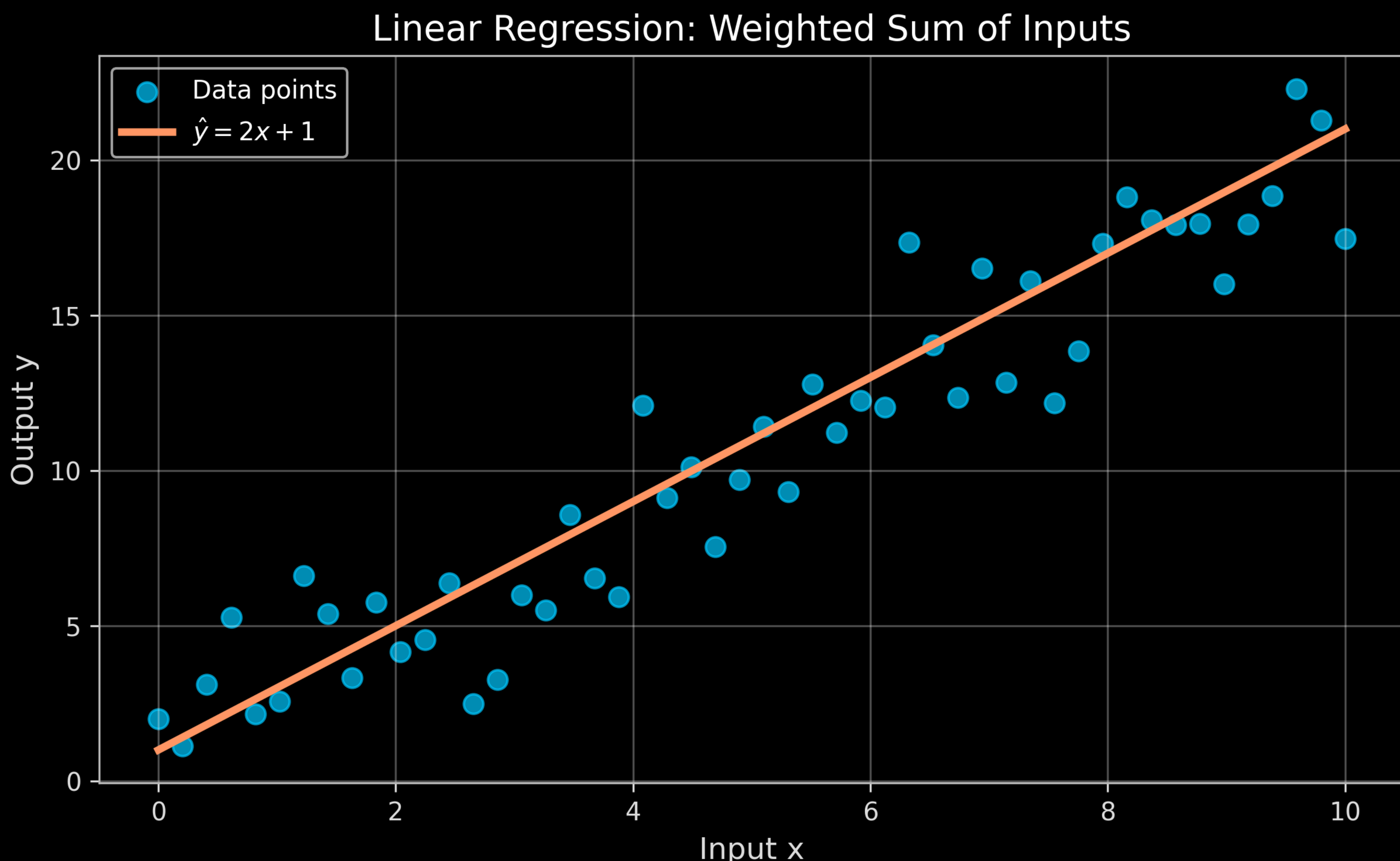
Or in vector notation:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b$$

Components:

- $\mathbf{x} = [x_1, x_2, \ldots, x_n]$: input features
- $\mathbf{w} = [w_1, w_2, \ldots, w_n]$: weights (learned)
- b: bias term (intercept)
- $\hat{y}$: predicted output

Visualizing Linear Regression



From Linear Regression to a Neuron

Linear Regression:

$$\hat{y} = \mathbf{w}^\top \mathbf{x} + b$$

- Output can be any real number
- Good for regression tasks
- Cannot model non-linear patterns

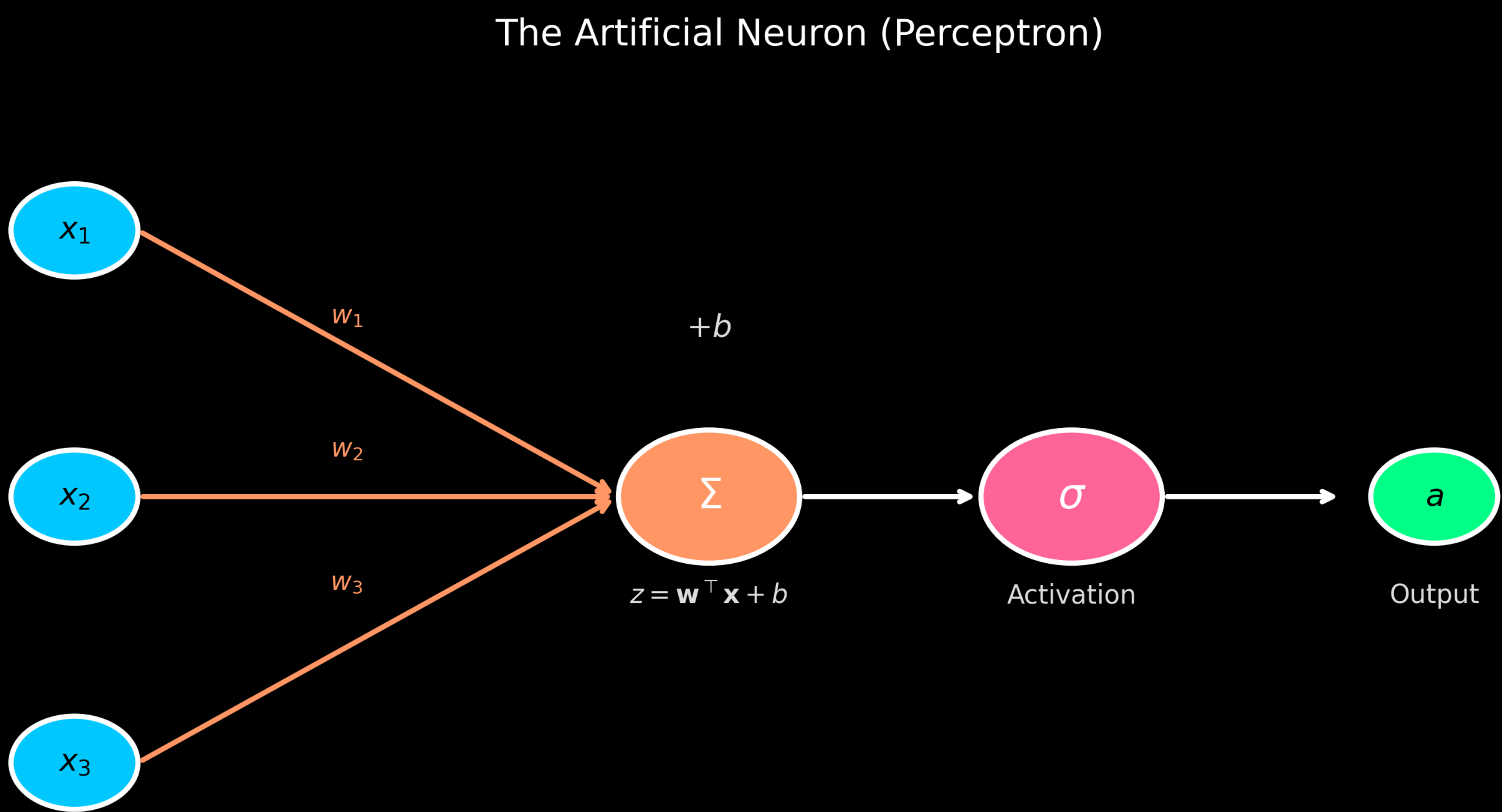
Single Neuron:

$$z = \mathbf{w}^\top \mathbf{x} + b$$

$$a = \sigma(z)$$

- Apply activation function σ
- Can introduce non-linearity
- Building block of neural networks

The Perceptron: First Artificial Neuron



Mathematical Formulation

Step 1: Compute the weighted sum (pre-activation)

$$z = \sum_{i=1}^{n} w_i x_i + b = \mathbf{w}^\top \mathbf{x} + b$$

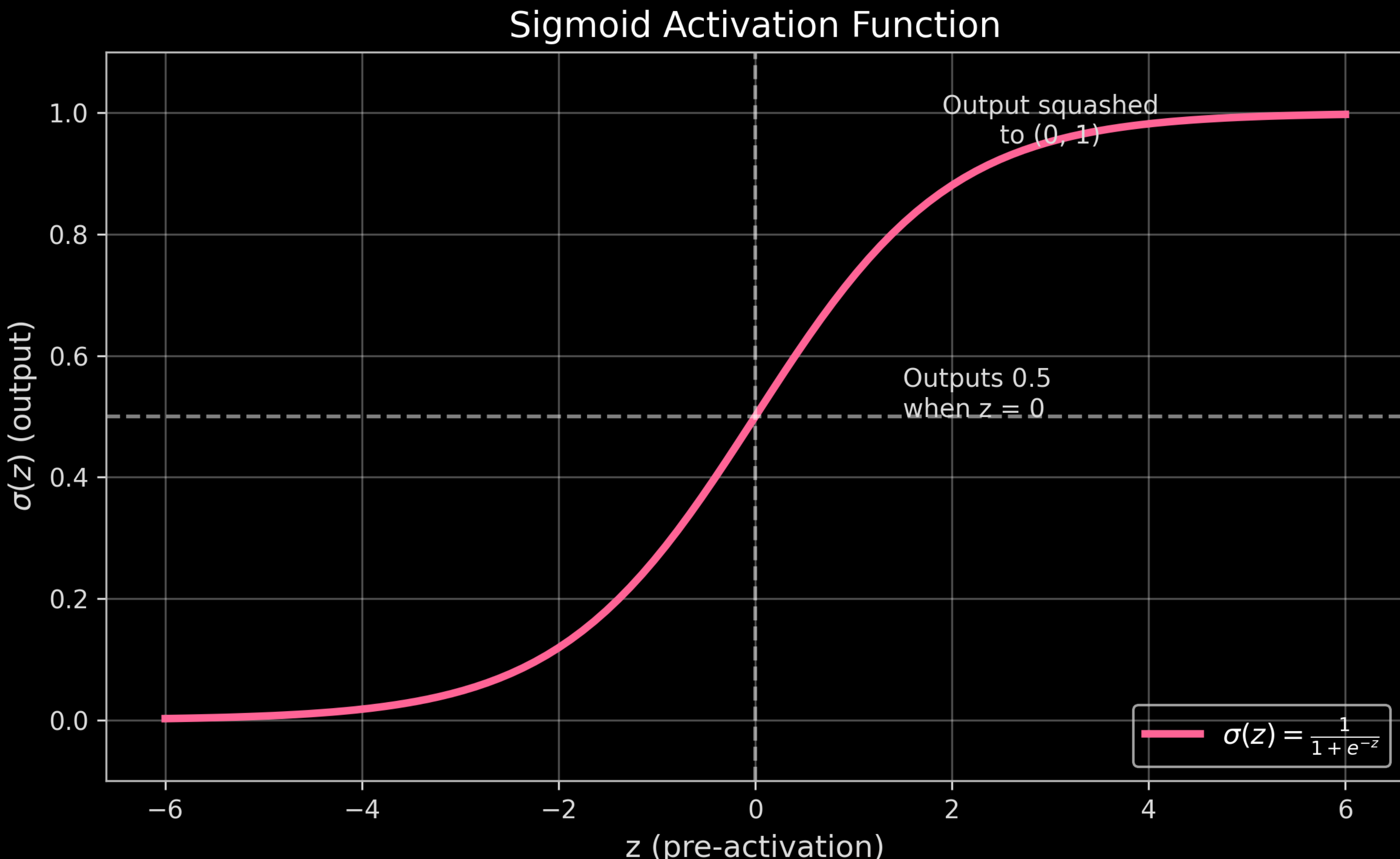
Step 2: Apply the activation function

$$a = \sigma(z)$$

Together:

$$a = \sigma\left(\mathbf{w}^\top \mathbf{x} + b\right)$$

The Sigmoid Activation Function



Connection to Logistic Regression

Logistic regression is a single neuron with sigmoid activation!

Logistic Regression:

$$P(y = 1|\mathbf{x}) = \frac{1}{1 + e^{-(\mathbf{w}^\top \mathbf{x} + b)}}$$

Single Neuron:

$$a = \sigma(\mathbf{w}^\top \mathbf{x} + b)$$

where $\sigma(z) = \frac{1}{1+e^{-z}}$

Key insight: You already know neural networks—logistic regression is the simplest one!

Why Neurons? The Limitation of Linear Models

What We've Covered

In this video, we've learned:

- Linear regression computes a weighted sum of inputs plus bias
- A neuron adds an activation function to this weighted sum
- The sigmoid function squashes outputs to (0, 1)
- Logistic regression is a single neuron with sigmoid activation
- Non-linear activation enables modeling complex patterns